Cumulativity from Homogeneity

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When two plural elements occur in a sentence, they give rise to the cumulative reading:

(1) a. The ten hens are sitting on the fifteen eggs.

b. **Cumulative readings**
Each of the hens is sitting on some eggs.
Each of eggs is covered by a hen.

On this particular example, it is not clear that there is anything semantically meaningful to say:

(2) "X *sits on* Y" if the totality of Y is covered by X

Cumulativity with quantifiers

(3) The ten hens are sitting on every one of the fifteen eggs.

Cumulativity with quantifiers

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$\neq$ for every one of the fifteen eggs x, the ten hens are sitting on x

Cumulativity with quantifiers

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Cumulativity with quantifiers

(3) The ten hens are sitting on every one of the fifteen eggs.
$\neq$ for every one of the fifteen eggs x, the ten hens are sitting on x
$\approx$ the ten hens are sitting on the fifteen eggs.
$\approx$ *Each of the hens is sitting on some eggs.*
Each of eggs is covered by a hen.

Cumulativity with quantifiers

Maybe this is just a special case of group reading of *every* (i.e. *every* behaves like a definite plural).

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Cumulativity with quantifiers

But also other quantifiers:

(5) The ten hens are sitting on less than 5 eggs.

Cumulativity with quantifiers

But also other quantifiers:

(5) The ten hens are sitting on less than 5 eggs.
$\neq$ for less than 5 eggs x, the ten hens are sitting on x

Meaning postulates are not enough! Cumulative readings should sometimes be derived compositionally (cf Beck et al., 2000)

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Many roads to compositional cumulativity

- Amend clause composition + meanings of quantifier
  - event semantics (Kratzer, 2000; Landman, 2012; Schein, 1993)
  - PCDRT (Brasoveanu, 2013)
  - Cross-categorial pluralities (Haslinger et al., 2018; Schmitt, 2013, 2015)
- Amend interpretation of plurals

  $\rightsquigarrow$ *grounded in properties of homogeneity*

Our main dataset:

(6) a. The ten hens are sitting on every one of the fifteen eggs.

b. The ten video-games taught every quarterback less than 8 plays. (Schein, 1993)

c. The ten hens are sitting on less than 5 eggs.

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(6) a. The ten hens are sitting on every one of the fifteen eggs.

b. The ten video-games taught every quarterback less than 8 plays. (Schein, 1993)

c. The ten hens are sitting on less than 5 eggs.

Left aside for the time being:

(7) The ten hens are sitting on the fifteen eggs.

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Homogeneity

(8) a. The dancers are smiling.

b. The dancers aren't smiling.

Homogeneity

(8) a. The dancers are smiling.

b. The dancers aren't smiling.

Neither (a) nor (b) is adequate when half of the dancers are smiling: truth-value gap!

Homogeneity

(Bar-Lev, 2018): underlying existential meanings for plurals

(9) a. The dancers aren't smiling.

b. $\neg\exists X \in$ dancers, X is smiling

Homogeneity

(Bar-Lev, 2018): underlying existential meanings for plurals + strengthening in positive contexts

(9) a. The dancers aren't smiling.

b. $\neg\exists X \in$ dancers, X is smiling

(10) a. The dancers are smiling.

b. $\textsc{Exh}\ \exists X \in$ dancers, X is smiling

Cumulativity (Bayer, 2013)

Interestingly, existential meanings for plural interpretation have also been proposed in the context of cumulativity.

(11) The ten chickens are sitting on every egg
$\neq$ for every egg x, the chickens are sitting on x
$\approx$ for every egg x, one or more chickens are sitting on x

Bird's eye view

Homogeneity (Bar-Lev, 2018)

- **Underlying Meaning:** $\exists$
- Truth-value gap

Cumulativity (Bayer, 2013)

- **Underlying Meaning:** $\exists$

Bird's eye view

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- **Underlying Meaning:** $\exists$
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- **Underlying Meaning:** $\exists$
- $\exists$ takes lowest scope in the clause (Schein, 1995)

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Cumulativity

(12) a. The ten hens are sitting on every egg

b. **Truth-conditions**
Every egg was sat on by one of the ten hens. (Bayer, 2013)

Cumulativity

(12) a. The ten hens are sitting on every egg

b. **Truth-conditions**

Every egg was sat on by one of the ten hens. (Bayer, 2013)

~~One hen sat on every egg~~ (*$\exists \gg \forall$, Schein, 1995)

Cumulativity

(12) a. The ten hens are sitting on every egg

b. **Truth-conditions**

Every egg was sat on by one of the ten hens. (Bayer, 2013)

Every hen sat on one or more eggs.

Cumulativity

(13) a. The ten hens aren't sitting on every egg.

b. **Truth-conditions**

not every egg was sat on by one (or more) hens.

c. **Unattested**

either not every egg was sat on by one (or more) hens.

or not every hen sat on one or more eggs.

Cumulativity

(14) a. The ten video-games taught every quarterback less than 5 plays.

b. **Truth-conditions**
Every quarterback was taught less than 5 plays by one or more video-games.

Cumulativity

(14) a. The ten video-games taught every quarterback less than 5 plays.

b. **Truth-conditions**
Every quarterback was taught less than 5 plays by one or more video-games.
~~*for every quarterback x,*~~
~~*some of the video-games taught x less than 5 plays*~~
(*$\forall \gg \exists \gg$ less than 5)

Cumulativity

(14) a. The ten video-games taught every quarterback less than 5 plays.

b. **Truth-conditions**
Every quarterback was taught less than 5 plays by one or more video-games.
Every video game taught a play to a quarterback.

Cumulativity

(15) a. The ten hens are sitting on less than 5 eggs.

b. **Truth-conditions**

Less than 5 eggs are sat on by one (or more) hens.

Every hen sat on one or more eggs.

Cumulativity

(16) a. The ten hens aren't sitting on less than 5 eggs.

b. **Truth-conditions**
At least 5 eggs are sat on by one (or more) hens.

Let's reflect on our dataset:

(17) *Every egg was sat on by one (or more) hens.* *(every $\gg \exists$)*

Every hen sat on one or more eggs.

(18) *Every quarterback was taught less than 5 plays by one or more video-games.* *(every $\gg$ less than 5 $\gg \exists$)*

Every video-game taught at least one play to a quarterback.

(19) *Less than 5 eggs are sat on by one (or more) hens.*

(less than 5 $\gg \exists$)

Every hen sat on one or more eggs.

Generalizations

- Bayer's paraphrase is adequate for negative sentences but is missing an inference for positive sentences
- The existential meaning that Bayer gives the plural must take lowest scope within the clause

$\rightsquigarrow$ *what about homogeneity?*

Homogeneity (Bar-Lev, 2018)

- **Underlying Meaning:** $\exists$
- $\exists$ takes lowest scope in the clause
- Additional inferences in positive sentences

Cumulativity (Bayer, 2013)

- **Underlying Meaning:** $\exists$
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Independent evidence for low-scoping existential

We need a plural-referring expression and a scope bearer

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We need a plural-referring expression and a scope bearer, *which is not one of the quantifiers that gives rise to cumulative readings!*

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(20) The ten dogs only bit Gabito.

Independent evidence for low-scoping existential

We need a plural-referring expression and a scope bearer, *which is not one of the quantifiers that gives rise to cumulative readings!*

(20) The ten dogs only bit Gabito.

a. **the ten dogs**$_{\exists}$ $\gg$ **only**
Some of the dogs bit no one beyond Gabito

b. **only** $\gg$ **the ten dogs**$_{\exists}$
No one other than Gabito was bit by any of the dogs

Independent evidence for low-scoping existential

We need a plural-referring expression and a scope bearer, *which is not one of the quantifiers that gives rise to cumulative readings!*

(20) The ten dogs only bit Gabito.

a. **the ten dogs**$_{\exists}$ $\gg$ **only**
Some of the dogs bit no one beyond Gabito

b. **only** $\gg$ **the ten dogs**$_{\exists}$
No one other than Gabito was bit by any of the dogs

(21) The ten dogs only bit Gabito.

(21) I doubt that the ten dogs only bit Gabito.

(21) I doubt that the ten dogs only bit Gabito.

a. **the ten dogs**$_{\exists}$ $\gg$ **only**
it's not the case that some of the dogs bit no one other than Gabito.

b. **only** $\gg$ **the ten dogs**$_{\exists}$
it's not the case that no one other than Gabito was bit by any of the dogs.

(21) I doubt that the ten dogs only bit Gabito.

a. **the ten dogs**$_{\exists}$ $\gg$ **only**
all of the dogs bit someone other than Gabito.

b. **only** $\gg$ **the ten dogs**$_{\exists}$
some of the dogs bit someone other than Gabito.

(21) I doubt that the ten dogs only bit Gabito.

a. **the ten dogs**$_{\exists}$ $\gg$ **only**
all of the dogs bit someone other than Gabito.

b. **only** $\gg$ **the ten dogs**$_{\exists}$
some of the dogs bit someone other than Gabito.

Independent evidence for low-scoping existential

Quantifiers in subject position in general have a hard time under-scoping VP *only*.

(22) a. A dog only bit Gabito. ($\exists \gg$ only)
$\neq$ *Only Gabito was bit by a dog*

b. Every dog only bit Gabito. ($\forall \gg$ only)
$\neq$ *Only Gabito was bit by every dog*

That reading is sometimes available with unaccented *some*

(23) *Sm* snow only fell on Gabito.

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Homogeneity + Cumulativity

- **Underlying Meaning:** $\exists$
- $\exists$ takes lowest scope in the clause
- Additional inferences in positive sentences

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Predicting low-scope

How to ensure existential interpretation for plurals and lowest-scope?

Predicting low-scope

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(24) The ten dancers $\exists$-Dist smiled (Bar-Lev, 2018)

Predicting low-scope

How to ensure existential interpretation for plurals and lowest-scope?

(24) a. The ten dancers $\exists$-smiled

b. $[\![\exists\text{-smiled}]\!] = \lambda X.\ \exists X' \prec X,\ X'$ smiled
$\approx$ some of X smiled

Many parallels:

(25) **Derived Kind Predication**

a. That kind of insect bit Mathilda.
$\approx$ *some insects of that kind*

b. That kind of insect only bit Mathilda.
(only $\gg \exists_{\text{instance}}$)

(26) **Metonymy**

a. IBM is in Paris.
$\approx$ *some branch of IBM*

b. IBM is only in Paris.
(only $\gg \exists_{\text{physical-branch}}$)

(27) a. The ten mosquitoes bit every hiker.

b. $[\![\exists\text{-bit}]\!]$
$= \lambda Y.\lambda X.\ \exists X' \prec X, \exists Y' \prec Y,\ \ X' \text{ bit } Y'$

c. **Attested truth-conditions:**
Every hiker was bit by one (or more) mosquitoes.
Every mosquito bit a hiker.

(27) a. The ten mosquitoes bit every hiker.

b. $[\![\exists\text{-bit every hiker}]\!]$
$= \lambda X.\ \forall y \in \text{hiker},\ \exists X' \prec X, \exists Y' \prec y,\ \ X \text{ bit } Y'$

c. **Attested truth-conditions:**
Every hiker was bit by one (or more) mosquitoes.
Every mosquito bit a hiker.

(27) a. The ten mosquitoes bit every hiker.

b. $[\![\exists\text{-bit every hiker}]\!]$
$= \lambda X.\ \forall y \in \text{hiker},\ \exists X' \prec X,\ \ X \text{ bit } y$

c. **Attested truth-conditions:**
Every hiker was bit by one (or more) mosquitoes.
Every mosquito bit a hiker.

(27) a. The ten mosquitoes bit every hiker.

b. $[\![$the ten mosquitoes $\exists$-bit every hiker$]\!]$
$= \forall y \in$ hiker, $\exists X' \prec \iota$10-mosquitoes, X bit y

c. **Attested truth-conditions:**
Every hiker was bit by one (or more) mosquitoes.
Every mosquito bit a hiker.

(28) a. The ten mosquitoes bit less than 8 hikers.

b. $[\![\exists\text{-bit}]\!]$
$= \lambda Y.\lambda X.\ \exists X' \prec X, \exists Y' \prec Y,\ \ X' \text{ bit } Y'$

c. **Attested truth-conditions:**
Less than 8 hikers were bit by one (or more) mosquitoes.
Every mosquito bit a hiker.

(28) a. The ten mosquitoes bit less than 8 hikers.

b. $[\![\exists\text{-bit less than 8 hikers}]\!]$
$= \lambda X.\ \#\{y \in \text{hiker} \mid \exists X' \prec X, \exists Y' \prec y,\ X \text{ bit } Y'\} < 8$

c. **Attested truth-conditions:**
Less than 8 hikers were bit by one (or more) mosquitoes.
Every mosquito bit a hiker.

(28) a. The ten mosquitoes bit less than 8 hikers.

b. $[\![\exists\text{-bit less than 8 hikers}]\!]$
$= \lambda X.\ \#\{y \in \text{hiker} \mid \exists X' \prec X,\ X \text{ bit } y\} < 8$

c. **Attested truth-conditions:**
Less than 8 hikers were bit by one (or more) mosquitoes.
Every mosquito bit a hiker.

(28) a. The ten mosquitoes bit less than 8 hikers.

b. $[\![$the ten mosquitoes $\exists$-bit less than 8 hikers$]\!]$
$= \#\{y \in \text{hiker} \mid \exists X' \prec \iota 10\text{-mosquitoes},\ X \text{ bit } y\} < 8$

c. **Attested truth-conditions:**
Less than 8 hikers were bit by one (or more) mosquitoes.
Every mosquito bit a hiker.

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(29) a. The ten dancers $\exists$-smiled.

b. $\exists X' \prec \iota$10-dancers, X' smiled

c. **Attested truth-conditions:**
Some dancer smiled.
Every dancer smiled.

(30) a. The ten mosquitoes bit every hiker.

b. $\forall y \in$ hiker, $\exists X' \prec \iota$10-dancers, X' bit y

c. **Attested truth-conditions:**
Every hiker was bit by one (or more) mosquitoes.
Every mosquito bit a hiker.

The needed strengthening mechanism must be:

- Polarity-sensitive
- Relevance-sensitive

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$\rightsquigarrow$ looks like pragmatic enrichment

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- Polarity-sensitive
- Relevance-sensitive

$\rightsquigarrow$ looks like pragmatic enrichment

What pragmatic enrichment?

(31) a. The ten dancers $\exists$-smiled.

b. $\exists X' \prec \iota$10-dancers, X' smiled

c. **Attested truth-conditions:**
Some dancer smiled.
Every dancer smiled.

(31) a. The ten dancers $\exists$-smiled.

b. $\exists X' \prec \iota 10$-dancers, X' smiled

c. **Attested truth-conditions:**
Some dancer smiled.
Every dancer smiled.

Bar-Lev (2018)'s diagnostic: free choice-like mechanism

(32) a. I don't like ANY dog.

b. I like ANY dog.

But it's difficult to see (33) as a FC inference

(33) a. The ten mosquitoes bit every hiker.

b. $\forall y \in \text{hiker}, \exists X' \prec \iota 10\text{-dancers},\ X' \text{ bit } y$

c. **Attested truth-conditions:**
Every hiker was bit by one (or more) mosquitoes.
Every mosquito bit a hiker.

(34) a. Mosquito 1 and Mosquito 2 bit every hiker.

b. $\forall y \in \text{hiker}, \exists X' \prec \text{M}_1 + \text{M}_2,\ X'$ bit y

c. **Attested truth-conditions:**
Every hiker was bit by Mosquito 1 or Mosquito 2.
Mosquito 1 bit a hiker.
Mosquito 2 bit a hiker.

(34) a. Mosquito 1 and Mosquito 2 bit every hiker.

b. $\forall y \in \text{hiker}, \text{M}_1 \text{ bit } y \vee \text{M}_2 \text{ bit } y$

c. **Attested truth-conditions:**
Every hiker was bit by Mosquito 1 or Mosquito 2.
Mosquito 1 bit a hiker.
Mosquito 2 bit a hiker.

$\rightsquigarrow$ *distributive implicatures!*

(35) a. Mosquito 1 and Mosquito 2 bit many hikers.

b. Many $y \in$ hiker, $\exists X' \prec \mathrm{M}_1 + \mathrm{M}_2,\ X'$ bit y

c. **Attested truth-conditions:**
Many hikers were bit by Mosquito 1 or Mosquito 2.
Mosquito 1 bit a hiker.
Mosquito 2 bit a hiker.

(35) a. Mosquito 1 and Mosquito 2 bit many hikers.

b. Many $y \in \text{hiker}, \text{M}_1 \text{ bit } y \vee \text{M}_2 \text{ bit } y$

c. **Attested truth-conditions:**
Many hikers were bit by Mosquito 1 or Mosquito 2.
Mosquito 1 bit a hiker.
Mosquito 2 bit a hiker.

$\rightsquigarrow$ *distributive implicatures!*

Non-cumulative sentences

(36) FC implicatures: $A(x) \vee B(x)$

a. $A(x)$

b. $B(x)$

Cumulative sentences

(37) Distributive implicatures: $Qx, A(x) \vee B(x)$

a. $\exists x, A(x)$

b. $\exists x, B(x)$

- Free Choice: recursive exhaustification (Fox, 2007)
- Distributive implicature: recursive exhaustification (Bar-Lev et al., 2016)

- Free Choice: recursive exhaustification (Fox, 2007)
- Distributive implicature: recursive exhaustification (Bar-Lev et al., 2016)

In ongoing work, Amir Anvari and I have suggested that these two inferences are part of much broader set of inferences, which are all derivable from the principle of Logical Integrity (Anvari, 2018).

I'll assume recursive exhaustification. But *any strengthening operation which delivers these inferences is principle available to us.*

Analysis

Assumptions

- A grammatical strengthening operation
- Sub-domain alternatives[1]:
  $[\![\exists\text{-bit}_{C,\,C'}]\!] = \lambda Y.\lambda X.\ \exists_C X' \prec X, \exists_{C'} Y' \prec Y,\ \ X' \text{ bit } Y'$
- Exhaustification is recursive: $\textsc{Exh}^2\ S := \textsc{Exh}\ \textsc{Exh}\ S$
- All quantifiers have *some* as an alternative.

[1] There is a simplification in the formula below ; as written here, collective interpretation will be excluded. Very similar to the candidate meanings of Križ et al. (2017)

Recap

Glossing over lengthy computations:

FC strengthening (Fox, 2007)

$\textsc{Exh}^2_C(\exists_C x, A(x)) = \forall x, A(x)$

Distributive strengthening (Bar-Lev et al., 2016)

For $Q \neq$ *no NP*,
$\textsc{Exh}^2_C(Qy, \exists_C x, A(x, y)) = (Qy, \exists x, A(x, y)) \wedge (\forall x, \exists y, A(x, y))$

Recap

Glossing over lengthy computations:

FC strengthening (Fox, 2007)

$\textsc{Exh}^2_C(\exists_C x, A(x)) = \forall x, A(x)$

Distributive strengthening (Bar-Lev et al., 2016)

For $Q \neq$ *no NP*,

$\textsc{Exh}^2_C(Qy, \exists_C x, A(x,y)) = (Qy, \exists x, A(x,y)) \wedge (\forall x, \exists y, A(x,y))$

$\textsc{Exh}^2_C(Q_1 y, Q_2 z, \exists_C x, A(x,y,z)) =$

$(Q_1 y, Q_2 z, \exists x, A(x,y,z)) \wedge (\forall x, \exists y, \exists z, A(x,y,z))$

(38) a. $\textsc{Exh}^2$ The ten dancers $\exists$-smiled$_C$.

b. $\textsc{Exh}^2(\exists X' \prec \iota 10\text{-dancers},\ X' \text{ smiled})$

c. **Attested truth-conditions:**
Some dancer smiled.
Every dancer smiled. (FC inference)

(38) a. $\textsc{Exh}^2$ The ten dancers $\exists$-smiled$_C$.

b. $\forall X' \prec \iota$10-dancers, X' smiled

c. **Attested truth-conditions:**
Some dancer smiled.
Every dancer smiled. (FC inference)

(39) a. $\textsc{Exh}^2$ The ten mosquitoes $\exists$-bit$_C$ every hiker.

b. $\textsc{Exh}^2\,(\forall y \in \text{hiker}, \exists_C X' \prec \iota 10\text{-mosquitoes},\ X' \text{ bit } y)$

c. **Attested truth-conditions:**
Every hiker was bit by one (or more) mosquitoes.
Every mosquito bit a hiker. (distributive implicature)

(39) a. $\textsc{Exh}^2$ The ten mosquitoes $\exists$-bit$_C$ every hiker.

b. $\forall y \in$ hiker, $\exists X' \prec \iota$10-mosquitoes, X' bit y z
$\forall x \prec$ 10-mosquitoes, $\exists x \in$ hiker, X' bit y

c. **Attested truth-conditions:**
Every hiker was bit by one (or more) mosquitoes.
Every mosquito bit a hiker. (distributive implicature)

(40) a. $\textsc{Exh}^2$ The ten video-games $\exists$-taught$_C$ every quarterback less than 8 plays.

b. $\textsc{Exh}^2(\forall y \in \text{quarterback}, < 8z \in \text{plays}, \exists_C X' \prec \iota 10\text{-video-games}, X' \text{ taught } y\ z)$

c. **Attested truth-conditions:**
Every quarterback was taught less than 8 plays by one (or more) video-games.
Every video-game taught a play to a quarterback.
(distributive implicature)

(40) a. $\textsc{Exh}^2$ The ten video-games $\exists$-taught$_C$ every quarterback less than 8 plays.

b. $\forall y \in$ quarterback, $< 8z \in$ plays, $\exists X' \prec$ ι10-video-games, X' taught y
$\forall x \prec \iota$10-mosquitoes, $\exists x \in$ quarterback, $\exists z \in$ play, x taught y z

c. **Attested truth-conditions:**
Every quarterback was taught less than 8 plays by one (or more) video-games.
Every video-game taught a play to a quarterback.
(distributive implicature)

Summary

Analysis

- **Underlying Meaning:** $\exists$
- $\exists$ takes lowest scope in the clause
  $\rightsquigarrow$ *built into the meaning of the verb*
- Additional inferences in positive sentences
  $\rightsquigarrow$ *strengthening as a variation inference*

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More complex cases

- What about ordinary cumulative sentences?
- Asymmetries in cumulative readings

Descriptively, ordinary cumulative sentences behave just like cumulative sentences with quantifiers.

(41) The mosquitoes bit the hikers.

a. **Truth-conditions:**
A mosquito bit a hiker.
Every mosquito bit a hiker.
Every hiker was bit a by a mosquito.

(42) a. The mosquitoes didn't bite the hikers.

b. **Truth-conditions:**
Not [a mosquito bit a hiker].

Analytically:

(43) a. not [the mosquitoes $\exists$-bit the hikers]

b. $\neg\exists x \prec \iota$mosquitoes, $\exists y \prec \iota$hikers, x bit y

c. **Attested truth-conditions:**
Not [a mosquito bit a hiker].

(44) a. $\textsc{Exh}^2_C$ the mosquitoes $\textsc{Exh}^2_{C'}$ $\exists$-bit$_{C,\,C'}$ the hikers

b. $[\![\exists\text{-bit}_{C,\,C'}$ the hikers$]\!]$
$= \lambda Y.\ \exists_{C'} x \prec \iota\text{hikers}, \exists_C y \prec Y,\ y$ bit x

c. **Attested truth-conditions:**
Every mosquito bit a hiker
Every hiker was bit by a mosquito

(44) a. $\textsc{Exh}^2_C$ the mosquitoes $\textsc{Exh}^2_{C'}$ $\exists$-bit$_{C,\,C'}$ the hikers

b. $[\![\textsc{Exh}^2\ \exists\text{-bit}_{C,\,C'}\ \text{the hikers}]\!]$
$= \lambda Y.\ \forall x \prec \iota\text{hikers},\ \exists_{C'} y \prec Y,\ y \text{ bit } x$ (FC strengthening)

c. **Attested truth-conditions:**
Every mosquito bit a hiker
Every hiker was bit by a mosquito

(44) a. $\textsc{Exh}^2_C$ the mosquitoes $\textsc{Exh}^2_{C'}$ $\exists$-bit$_{C,\,C'}$ the hikers

b. $[\![$the mosquitoes $\textsc{Exh}^2$ $\exists$-bit$_{C,\,C'}$ the hikers$]\!]$
$= \forall x \prec \iota$hikers$, \exists_{C'} y \prec \iota$mosquitoes, y bit x

c. **Attested truth-conditions:**
Every mosquito bit a hiker
Every hiker was bit by a mosquito

(44) a. $\textsc{Exh}^2_C$ the mosquitoes $\textsc{Exh}^2_{C'}$ $\exists$-bit$_{C,\,C'}$ the hikers

b. $[\![\textsc{Exh}^2$ the mosquitoes $\textsc{Exh}^2$ $\exists$-bit$_{C,\,C'}$ the hikers$]\!]$
$= \forall x \prec \iota$hikers, $\exists y \prec \iota$mosquitoes, y bit x
$\wedge \forall y \prec \iota$mosquitoes, $\exists x \prec \iota$hiker, y bit x
(distributive strengthening)

c. **Attested truth-conditions:**
Every mosquito bit a hiker
Every hiker was bit by a mosquito

Table of contents

What is cumulativity?

Descriptively...

- Parallel between cumulativity and homogeneity
- Cumulativity : low-scoping existentials and gaps
- Homogeneity: low-scoping existentials and gaps

Analysis

- Predicting low-scope
- Becoming stronger via variation inferences

More complex cases

- What about ordinary cumulative sentences?
- Asymmetries in cumulative readings

Asymmetries in cumulative readings of every

(45) a. Every mosquito bit the hikers. (#cumulative)
b. The mosquitoes bit every hiker. (cumulative)

The asymmetry is an asymmetry of c-command (Champollion, 2010; Haslinger et al., 2018).

The two sentences are predicted to have similar negative meanings *mutatis mutandis*. And this seems right.

(46) a. Not every mosquito bit the hikers. (#cumulative)

b. $[\![(46a)]\!] = \neg\forall h \in \text{mosquito}, \exists y \prec \iota\text{hikers}, m \text{ bit } h$

c. **Attested TC:** some mosquitoes bit no hikers (Križ et al., 2015)

(47) a. The mosquitoes didn't bite every hiker. (cumulative)

b. $[\![(47a)]\!] = \neg\forall h \in \text{hiker}, \exists y \prec \iota\text{mosquitoes}, m \text{ bit } h$

c. **Attested TCs:** not every hiker was bit by a mosquito

So the difference must stem from the position of $\textsc{Exh}$.

(48) a. Every mosquito $\textsc{Exh}^2_C$ the hikers $\exists$-bit$_C$ (#cumulative)

b. $[\![(48a)]\!] = \forall h \in$ mosquito, $\textsc{Exh}^2_C\ \exists_C y \prec \iota$hikers, m bit h

c. **Attested TC:**

Every mosquito bit a hiker.

Every mosquito bit every hiker. (Križ et al., 2015)

(49) a. $\textsc{Exh}^2_C$ The mosquitoes bit every hiker. (cumulative)

b. $[\![(49a)]\!] = \textsc{Exh}^2_C\ \forall h \in$ hiker, $\exists_C y \prec \iota$mosquitoes, m bit h

c. **Attested TC:**

Every mosquito was bit by a hiker.

Every mosquito bit a hiker. (Križ et al., 2015)

Syntactic stipulation

$\textsc{Exh}^2_C$ applies at the position that the corresponding DP is merged.

Homogeneity + Cumulativity

- **Underlying Meaning:** $\exists$
- $\exists$ takes lowest scope in the clause
- FC+Distributive implicatures as a mechanism for strengthening in positive contexts

Thank you!

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Collective predication

(50) $[\![\exists\text{-smiled}_C]\!] = \lambda X.\ \exists_C x \prec X, \exists X' \prec X,\ X'$ smiled

Cf. candidate meanings in Križ et al., 2017.